

ICMI Study 16

Challenging Mathematics in and beyond the Classroom

Proposal

Viktor Freiman, professor
Université de Moncton
Pavillon Jeanne-de-Valois
Moncton, New Brunswick
Canada E1A 3E9
E-mail: freimanv@umoncton.ca

Nancy Vézina, professor
University of Ottawa
145, Jean-Jacques-Lussier
Ottawa, Ontario
Canada K1N 6N5
E-mail: nancy.vezina@uottawa.ca

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Challenging Virtual Mathematical Environments: The Case of the CAMI Project

1. Introduction

It is well known that the development of problem solving abilities is an important part in the growth of mathematical understanding. Recent work (Lampert & Cobb, 2003; Sfard, 2003; 2001) has also revealed that the role of communication in the learning process is essential to construct a deeper understanding of mathematics concepts. “When students are given the opportunity to communicate about mathematics, they engage thinking skills and processes that are crucial in developing mathematical literacy” (Pugalee, 2001, p. 296). Thus communicating mathematical ideas becomes an important learning and teaching task to help students make sense of mathematics through authentic communication activities. The use of information and communication technologies (ICT) can become an important tool to create such significant activities. As a matter of fact, the *National Council of Teachers of Mathematics* (2000) believes that technology is an essential component to the learning and teaching of mathematics. Not only can technology guide the mathematical concepts that are being taught but it can also increase the learning opportunities for students. In fact, the following three assumptions led us to the development of the CAMI project: challenging problem solving is an essential part of learning mathematics, communication is a significant part of the learning of mathematics, and there is a meaningful use of ICT.

The Internet-based project CAMI (www.umoncton.ca/cami) is one of the didactical resources that have been developed to motivate school students K-12 to solve challenging problems, and thus contribute to the deepening of their understanding of mathematical concepts and the improvement of their abilities to reason and to communicate mathematically. Pursuing a long-time tradition of mathematicians to challenge each other with hard problems by exchanging with letters using mail, it turned to be a valuable example of meaningful integration of technology in a mathematics classroom of the 21st century. The fruit of a close collaboration between the university (Université de Moncton, professor Nancy Vézina, now at the University of Ottawa) and the school district (District scolaire 1, NB, director of Education Maurice Langlais), the project has three original goals: 1) be a tool to help francophone schoolchildren develop problem-solving and communication abilities; 2) be a tool for pre-service teacher training in math education helping students to learn how to integrate technology in a math classroom, how to evaluate a solution of mathematical problem in a formative way, and how to understand children’s mathematical reasoning and ideas beyond it; 3) be a resource for teacher in the school system. (Vézina & Langlais, 2002).

2. Theoretical framework

The last five years were marked by a profound shift of New Brunswick’s francophone school system toward new teaching and learning paradigms along with other Canadian provinces. Under strong influence of NCTM standards from one side, and new didactical approaches based on competencies from another side, the new mathematics curriculum puts emphasis on communication, use of technology and critical thinking in a genuine problem-solving environment (MENB, 2003). Aiming to meet the educational needs of all students, a new reform tries to find an appropriate didactical response to different challenges of the modern mathematics classroom.

Several studies conducted in the past five years showed the importance of giving young children access to powerful mathematical ideas (Perry & Dockett, 2002; Solomon, 2001). These studies stress that the student’s knowledge of mathematics is formed of internal networks of representations. The understanding of mathematics requires these representations be interlinked through structured coherent networks (Hiebert & Carpenter, 1992). This understanding happens when the learners face a meaningful problem allowing them to build necessary links and connections making sense of mathematical knowledge. Therefore, it is not surprising that the new curriculum in New Brunswick gives a central place to problem solving. It helps schoolchildren to get access to mathematical concepts and develop necessary thinking tools in order to realize their full potential (MENB, 2003).

In the new didactical paradigm called socio-constructivism, the teaching and the learning are not considered anymore as two distinct processes but as one complex process of teaching/learning in which the role of two parties (teacher – student) is essentially modified. Thus, in this process, the teacher and the student share the responsibility of interactions on knowledge inside the problem situations (*situations-problèmes*) proposed by the teacher (Jonnaert & Vander Borgh, 2003). At the same time, the teaching task becomes more complex and illustrates the paradox of devolution of situations (Brousseau, 1998) in which the teacher is obliged to teach everything necessary about the knowledge, has to tell the students how to solve the problems, without giving them a chance to make choices, trials and errors in order to modify

their own knowledge. Thus, the teachers risk missing their chances to ensure learning results for their pupils. Moreover, to access the superior level of their understanding of mathematics, the students need to reorganize their knowledge. But we cannot just tell the students how to reorganize their knowledge but also engage them in new situation problem, preparing ourselves for any kind of difficulties and obstacles and helping the students to overcome them (Sierpinska, 1994).

In her model for the development of mathematically promising children, Sheffield (1999) opted for the use of multi-dimension tasks within the scope of heuristic and open model which would contribute to the development of abilities to create, make links, investigate, communicate, and evaluate. Briefly saying, in order to succeed in teaching mathematics, it is necessary to create a good equilibrium between the routine and well-structured tasks, and those more creative and innovative. The students need to create a solid repertoire of positive experiences in problem solving, which would allow them to develop their self-confidence and potential (Klein, 2003).

While some children are struggling with mathematics, others might loose interest in mathematics because they are not challenged enough (Tempest, 1974). At the same time, the research shows that all children are eager to learn more and are able to learn more. One of our study conducted in a Montreal-based private elementary school, proved to be another case showing that an appropriate challenge can, not only foster a development of mathematical potential in gifted students in early grades, but it could also be beneficial for all students (Freiman, 2003; 2004). Many authors demonstrate a need for more challenging curriculum not only for the mathematically talented, but for a larger population of schoolchildren. Sheffield (1999) calls these children 'mathematically promising' and argues that a challenging program may also introduce them to the joys and frustrations of thinking deeply about a wide range of original, open-ended, or complex problems that encourage them to respond creatively in ways that are original, fluent, flexible, and elegant (Sheffield, 1999).

Several recent reflective studies of new teaching and learning virtual environments that started to appear on the wave of growing technology used in schools report a positive effect on the pupils' motivation toward mathematics. For example, the NRIC Project (www.nrich.math.org) was created in 1996 with the original idea to provide the brightest students with the enriched resources on-line that are not available otherwise in schools. In their framework for Mathematical Enrichment, the authors of the project share their research findings: 1) such on-line resources are not suitable solely for the most able but have something to offer pupils of nearly all abilities, 2) enrichment is not only an issue of content but a teaching approach that offers opportunities for exploration, discovery and communication, and 3) effective mediation offers a key to unlock the barriers to engagement and learning (Piggot, 2004). An example of pedagogically powerful virtual environment has been created within the project MathForum (mathforum.org). It is built on the idea of interaction between members of a virtual community that interact around the services and resources participants generate together. These interactions provide a basis for participant knowledge building about mathematics, pedagogy, and/or technology. The interactions also contribute to what might be described as a MathForum culture that encourages collaboration on problem posing and problem solving (Renninger & Shumar, 2002).

Seeing a problem-solving communication process as a teaching approach brings yet another dimension to such projects. The project of Solving Mathematical Problems within a Virtual Environment has been started by a group of educators from several Quebec universities (Charbonneau, 2000). It promotes a connection between teacher training programs and schools, setting up a virtual bank of mathematical problems and a computer environment for discussion and communication. The CAMI virtual learning community can be seen as another example of fruitful collaboration between schools and university that promotes changes in motivation and attitudes toward mathematics through the use of problem solving, communication and the use of technology. In the next section, we will examine closely the structure of the CAMI project as a virtual community, as a bank of challenging problems and as a user-friendly computer environment. We will also discuss a research context that is being set up in order to measure the impact of CAMI on different categories of participants.

3. What is the CAMI Project?

The functioning of the CAMI is quite simple. For 30 weeks of the school year, four problems are posted on the website every Monday morning. Problems are divided in the following matter: one problem intended for students at the primary level (Kindergarten to grade 2), one problem for students at the elementary level (grade 3 to grade 5), one problem for students at the middle school level (grade 6 to grade 8), and, finally, one problem for students at the high school level (grade 9 to grade 12). Students have the week to send in their solutions, which should include sufficient explanation to meet the goal of communicating mathematical information. Their solutions are submitted using a Web-based form (see Figure 2), or by sending an e-mail. Once the solutions are received, the pre-service teachers' work begins.

Working in teams, the pre-service teachers have to assess the work that was done by the students. They first look at the problem itself trying to anticipate different strategies and solutions, and to elaborate specific criteria of evaluation. They then do a formative assessment writing a personal comment to each student giving them constructive recommendations about how to improve their problem solving abilities. The messages are personally sent to each student that gave an e-mail address to receive feedback. They also pick, from all the solutions received, some that can be defined as exemplary to be featured on the website. Some pre-service teachers will also occasionally be involved in creating problems for the website. The work in the project gives the pre-service teachers a taste of real teaching. It helps them to better understand students' mathematical thinking and thus creates a positive dynamic between the pre-service teachers and their future students. At the same time, pre-service teachers improve their own problem solving and communication abilities, as well as teaching and assessment strategies.

The CAMI virtual community is built upon few basic principles:

- Friendly welcome: Every student from Kindergarten to Grade 12 is welcome to join the community at any time; it is free of charge and obligation.
- Math challenge: Everyone can submit a solution to any challenging math problem and send it to our CAMI team.
- Formative feedback: School students who send their solution get a personal comment from a university student; this comment is always positive and encouraging; it aims to encourage each participant to be persistent and continue to participate.
- Variety: The CAMI environment is open to a variety of styles and strategies; we try to understand different ways of thinking.
- Open communication: Communication is a vehicle of the community as it promotes knowledge sharing and knowledge building through collaboration and discussion.

4. CAMI: A user-friendly computer environment

The structure of the Internet site is very simple (see Figure 1.). The starting page gives a short presentation of the project. Three main menu items invite you to solve one of four problems of the week (*Problème de la semaine*), to read the analysis of the previous problem (*Solution du dernier problème*), and to search in the archive (*Problèmes archivés*). For students, there is also a possibility to play challenging games through several links to other sites (*Jeux et liens*), for teachers there is a teacher corner (*Information pour les enseignantes et enseignants*). One of the last two items directs to the list of all contributors to the project (team members, university students, consultants, programmers and sponsors (*Remerciements*), another provides visitors with the necessary technical information (*Information technique*). The same menu appears on all the screens in the same place making navigation easy and user-friendly.



Figure 1. Interface of the CAMI Website (www.umoncton.ca/cami)

5. Creating and solving challenging problems

In our choice of challenging problems for the CAMI database, we follow several criteria preferring the use of open-ended problems that are often imbedded in a real-life context (problem situation) with a challenging mathematical component allowing participants to construct mathematical knowledge through communication. The problems of the *Apprenti* category (Kindergarten to grade 2) are authentic. The problems for the other three categories (*Technicien*, *Ingénieur*, and *Expert*) are chosen with a friendly permission from the *MathForum* from their Problems of the Week (PoW) bank, translated and adapted to the New Brunswick curriculum. The choice of problems is made in order to cover a variety of topics: numbers and operations, patterns and relationships, shapes and space, statistics and probability – four main content domains of the New Brunswick K-12 math curriculum. Figure 2 shows an example from the *Apprenti* category.

134. Jus d'orange frais!
 Jessica prépare du jus d'orange frais à tous les matins. Habituellement, deux grosses oranges lui permettent de faire un petit verre de jus. Si elle prends plutôt 4 grosses oranges, elle peut obtenir un grand verre de jus.

La mère de Jessica a acheté à l'épicerie une douzaine d'oranges. Elle n'a cependant pas fait attention et elle a mis dans son sac 10 grosses oranges et deux petites. Combien de verres de jus d'orange Jessica peut-elle faire avec les oranges que sa mère a achetées?

N'oublie pas d'expliquer ta démarche et ta solution à l'aide de phrases complètes.

Pour les as : Peux-tu trouver plus d'une solution à ce problème?



Figure 2. A problem from the Apprenti category¹

The problem has several of the features mentioned above. It is accessible to very young children due to the small numbers, but might be a good challenge to the older ones. It uses a real-life context familiar to all students. It is open to various interpretations and different strategies. It is related to the school curriculum (numbers and operations, patterns and relationships) setting challenges that go beyond usual school routine. Figure 3 illustrates three different approaches children used to meet the challenge presented by the problem in Figure 2.

<i>Si je prends 2 grosses oranges pour faire un petit verre de jus et 4 grosses oranges pour en faire un grand verre, alors, je vais faire 5 petits verres de jus, en pressant 2 oranges à la fois. Ce qui est de mes 2 petites oranges je les mangerais ou, je me ferais un mini-verre de jus.</i> <i>Joannie</i>	If I take 2 big oranges to make a small glass of jus and 4 big oranges to make a big one, so I will make 5 glasses of jus, squeezing 2 oranges at the time. As for my 2 small oranges, I will eat them or I will very small glass of jus.
<i>Si 4 grosses oranges = 1 verre de jus O = 1 grosse orange o = 1 petite orange OOOOOOOOOOoo Prends 4 grosses pour 1 verre, il reste 6 grosses et 2o. Prends les autres 4 grosses, ce qui fait un autre verre, il reste 2 grosses et 2 petites. Prends 2 grosses, ce qui fait un demi verre, et 2 petites font 1 quart. R: Elle peut faire 2 verres et 3 quarts.</i> <i>Samuel</i>	If 4 big oranges = 1 glass of jus O = 1 big orange o = 1 small orange OOOOOOOOOOoo Take 4 big ones for 1 glass, you're left with 6 big et 2o. Take the other 4 big, it will give another glass, and you will be left with 2 big and 2 small. Take 2 big, which gives half a glass, and 2 small a quarter of a glass. A: She can make 2 glasses and three quarters.

¹ Translation: Every morning, Jessica prepares fresh orange juice for the whole family. Usually, 2 big oranges are enough to make a small glass of juice. With 4 big oranges, Jessica could make a big glass of juice. Jessica's mother has bought a dozen of oranges at the local market. Being a bit distracted that day, she took only 10 big oranges while two others were small ones. How many glasses of juice can Jessica make with these 12 oranges? Bonus: Can you find more than one solution?

<i>Bonjour!</i> <i>Moi, je crois qu'avec 10 grosses oranges et 2 petites oranges, Jessica peut faire 1 demi verre de jus d'orange et 5 autres petits verres.</i> <i>Pour les as: oui, je crois avoir une autre réponse. Je crois que Jessica peut aussi faire un petit verre de jus d'orange et deux autres gros verres, parce que, si on met deux petites oranges, je crois que ça fait une grande; on peut faire une demie verre, mais il va rester une grosse orange, mais moi, je me dis que deux petites oranges et une grosse orange font deux grosses oranges. Ma réponse est que Jessica peut faire un petit verre et deux gros verres.</i> <i>Marie-Eve</i>	Hi! Me, I think that with 10 big oranges et 2 small oranges, Jessica can make 1 half of a glass of orange jus and 5 small ones. Bonus: Yes, I think I have another answer. I think that Jessica can also make one small glass of jus and two other big glasses, because, if we put two small oranges, I think it will make one big, so we can make half of a glass, but we would be left with one big orange, but me, I said to myself that two little oranges and one big orange count as two big oranges. My answer is that Jessica can make one small glass and two big glasses.
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Figure 3. Examples of children's solution to the jus problem

6. CAMI: A research context

The research component of CAMI was not pre-dominant in the first years of its existence. The authors of the project conceived it just as an additional resource in mathematics education for teachers, university students and schoolchildren. In time we realized that the project went beyond its original goals and needs to be more closely examined from the researchers point of view.

According to the E-stat visitor monitoring system (www.estat.com), the number of visits to the CAMI site constantly grows, as does the number of solutions submitted electronically to our database. The geography of participants is also expanding, new schools from New Brunswick and other Canadian provinces discover CAMI and join in the community. It is difficult to evaluate the exact number of CAMI users because many schools use the site without sending solutions. We know that some solutions come from home (they arrive when the schools are closed), but we do not know whether it was an initiative of parents, of children themselves, or if it was a teacher's assignment for homework.

Our conversation with teachers and school administrators helps us to learn the existence of mini-CAMI communities when teachers give problems to students via Internet and children send their solution to the teachers e-mail address. Some teachers make children invent their own problems, post them on the school WEB site and organize a peer evaluation of solutions. One teacher told us that CAMI is her main resource for meaningful problem situations so she constructs her whole teaching based on CAMI problems that allow her to cover all curriculums.

The CAMI might also become a part of outreach mathematical activities. For example, The New Brunswick School District 9 is organizing once a year a Mathematical night bringing together approximately 150 youngsters from a dozen local schools to spend a whole night in various mathematics competitions, and CAMI is part of these activities. In fact, the phenomenal growing popularity of the project makes CAMI a more attractive object of research.

It is well known that projects like CAMI have a particular place in educational research because of their complexity and lack of appropriate research tools. We are only in the exploratory phase of the research trying to develop an appropriate theoretical framework and methodology. In our work, we are inspired by new paradigms in educational research such as action research, collaborative research and research development (see, for example, Anadon, 2001). In this article, we are focusing mostly on the description of how the CAMI works and analyze feedback from two groups of participants, schoolchildren and university students with the help of questionnaires.

We try to answer the question of what role the CAMI plays for both groups of participants and what are possible advantages and disadvantages of such a project. In the next section, we will present preliminary results of our exploratory study conducted in 2004-2005 in order to understand the impact of CAMI on both school and university groups of participants.

7. Results from the survey

The survey conducted in March-May 2004 has been separated in two parts. First, we elaborated and put on our site a survey for schoolchildren. Secondly, we conducted a survey among university students enrolled in our math education courses. Both surveys were anonymous. The participation of schoolchildren was voluntary; they could fill in an electronic form and submit it to us. At the same time, we asked all university students (from the K-8 program) who were present during the class to answer the questionnaires that were distributed in a paper form. Students had about 20 minutes to answer it. We got overall 96 responses from children and 66 from university students. In this section, we will share some of our findings from the survey that help to get insight into the role of CAMI for both groups of participants, schoolchildren and university students.

7.1 Findings from children's responses

A survey conducted with one hundred francophone students using the CAMI site allowed us to collect data presented in this analysis. The survey took place in April 2004. We evaluated around 96 participants of which 63% were girls. The age of the participants varied between 8 and 14+ years old; 16% of the participants said they were between 8 and 10, and 56% between 11 and 13, and 28% said they were 14 and older.

Most of the respondents were doing problems from categories *Apprenti* and *Technicien* (69%). To the question "What do you think about the problems of the CAMI site?" 27% of the students answered "Very interesting," 52% answered "Quite interesting," and 10% answered "Very annoying." Thus, in general, the students appreciate the idea of the CAMI site and find its content interesting and at the right difficulty level. Also, 47% of them find the interface of the site "Attractive" and 48% "Quite attractive."

To the question "How did suggestions sent in messages help you to improve your problem-solving strategies?" 19% of the students answered "Very much," 40% answered "Somewhat," 33% answered "A little bit," and 8% answered "Not at all." Regarding the question "The use of CAMI site, did it help you to improve your abilities in problem-solving?" 20% of the students answered "Very much," 46% answered "Somewhat," 26% answered "A little bit," and 8% answered "Not at all."

These results allow us to answer favorably to the first question, which was to determine if the CAMI site helps to improve pupils' abilities to solve mathematical problems and to increase their interest in mathematics. Noting that 88% of schoolchildren said that they would return to the CAMI next year.

In an open question about CAMI, 27 of 96 schoolchildren gave an explanation of what they like in CAMI. Their answers confirm a positive attitude toward CAMI. Children are saying that they are good, interesting, amusing and enjoyable problems (8 children). The work with our problems helps them to learn how to solve problems and to improve math skills (2 answers). One child mentioned a good variety of problems. Four respondents like computer environment saying that CAMI is a good site, with nice colors; it allows them to use computers and Internet (4 answers). Three respondents share their joy about CAMI (amusement, pleasure, fun), one child says that the work with CAMI makes a math class more interesting. Four children said they like games on the CAMI site. Finally, two more children said that everything on CAMI is good.

7.2 Findings from the university students' responses

One of the important questions of this study was to verify if the site CAMI improves pre-service teachers' understanding of students' needs. At least we can say that the majority of the students who participated in the survey recognized the positive impact and importance of the site for their education. The following are comments from students regarding the impact of CAMI on their learning:

- ☐ I very much like this site for students. They can work by themselves and be valorized at the same time. In the classroom, the problems of the site will be able to serve as a "*mise en situation*" to introduce certain concepts.
- ☐ Allows a teacher to incorporate ICT in mathematics.
- ☐ Motivates students in their learning.
- ☐ I think it allows the students to develop their abilities in mathematics, pushes the good students to go further, and a retroaction can help weaker students.
- ☐ Shows a tool that we will be able to use in our classroom.
- ☐ Allows thinking about students' mathematical reasoning.

Let's look closer into the data from the closed questions. The first group of questions is related to the opinion of what this project gave to the student in term of the pre-service teachers training program in math education.

The data from Table 1 show that, in general, the university students agree (strongly or partly) that the project is useful for their math education training. If we combine both 'agree' columns, the total will vary from 73% (improving math knowledge) to 97% (process of problem solving). The only item that falls below this interval is related to the question of whether the project helps to understand the new curriculum (41%). When we look at the first column (strongly agree), we can see that agreement with the better understanding of the importance of communication (75%) is significantly higher than other scores. The next items of this kind would be formative evaluation (62%) and understanding of schoolchildren's reasoning (50%). It is worth mentioning that 90% agree that we should continue to integrate the project into math education courses (56% strongly and 34% partly).

Table 1: Results from the university students' questionnaire (impact on their training)

Items	% Strongly agree (n = 66)	% Partly agree (n = 66)	% Partly disagree (n = 66)	% Strongly disagree (n = 66)	% Do no know (n = 66)
The project helps to understand the child's reasoning	50	47	3	0	0
The project helps to understand the process of problem solving	35	62	3	0	0
The project helps to understand the importance of communication in mathematics	73	24	3	0	0
The project helps to understand the new curriculum in New Brunswick	15	26	46	11	2
The project helps to improve my technique of formative evaluation in mathematics	62	32	6	0	0
The project allowed to review mathematical notions	45	27	13	3	2
The project allowed to improve my math knowledge	39	34	19	5	3
The project gave the idea of using computers in the mathematics classroom	41	44	11	2	2
It is worth it to continue to integrate the project in math education courses	56	34	8	0	2

The second part of the questionnaire was asking university students' opinion of what the project would give to the schoolchildren. All the answers in this category show that university students agree (strongly or partly) that the project is beneficial to schoolchildren. With a combined score varying from 79% (creativity) to 98% (value of personal comment), they affirm that it increases motivation in mathematics in general and in problem solving in particular. The project helps to develop mathematical communication and creativity, improve mathematical knowledge and reinforce computer skills. It also strengthens interdisciplinary links. If we look closer at some specific results, we can state that 79% of university students think that their work of writing a personal comment is important for schoolchildren. The next popular item is about developing computer skills (65%), competence in communication (61%) and interdisciplinary links (55%).

We also looked at students' opinion related to a category of schoolchildren for which the project would be the most beneficial (they had to select appropriate categories). Upon students' choice, the children that might benefit most from the project are mathematically able students (44 of 66 answers), next come children having difficulties (31 of 66), and slow-working kids (25 of 66).

We also considered students' opinion of what the project gives to the teachers. The data show that university students agree (mostly strongly) with percentages varying from 92% to 97% (for strongly agree – from 66% to 76%) on all items but one (following progress of schoolchildren). This remaining item shows rather reserved approval (72% for combined scores, but only 29% strongly agree).

The last portion of items is related to the content of the site and to its functioning. From these data, we learn that the structure, content and functioning of the Internet site CAMI is well appreciated by the university students. With combined scores varying from 75% to 95%, they approve strongly its facility to use (79%), functioning (77%) and attractiveness (68%). Regarding the content, there is a significant number of students that strongly agree that the site offers a good variety of problems (66%), the problems are affordable to children (62%), and the content corresponds to the curriculum (53%) and allows enriching it (43%). The item regarding the content reveals an important number of students that hesitate to evaluate them (8 to 15% choose the do not know option). Answering the question of how to develop the site, 65% think that we should leave it like it is, 26% think that we should modify it, 9% do not know or hesitate to answer.

8. Discussion and conclusions

In our preliminary analysis of the data, our attention is drawn to the fact that the project CAMI does, indeed, provide a challenging opportunity in mathematical communication for all its participants. Our example of an *Apprenti* problem shows that within a real-life context, different mathematical structures and relationships might be discovered by schoolchildren. The open-ended character of the problem situation allows children to choose different strategies and different ways of communication. University students who evaluate solutions in a formative way giving a personal feedback to each participant have approved these findings.

In fact, the importance of challenge in mathematics through problem solving and communication turns out to be an important discovery pre-service teachers make while doing a CAMI project within mathematics education courses. This evidence is confirmed by schoolchildren who say (although in less numbers) that this feedback helps them to improve their problem-solving skills.

Our findings show also that the university students find CAMI beneficial for the development of mathematical communication in young children. This communication is fruitful when it is bidirectional. This means that pupils are encouraged to communicate their mathematical ideas using a variety of tools. Pre-service teachers should be able to understand this variety, to appreciate it and to guide children through their problem-solving process. As survey results show, the majority of university students see the CAMI project as a valuable tool to penetrate a child's way of thinking. From another side, it is necessary to be objective and be able to challenge children's reasoning when it's incorrect and guide a child through the process of improvement. At the same time, university students seem to be united in the opinion that the project helps them learn how to understand a child's reasoning, to be open to a variety of strategies and tools of communication that children use, and provide a child with a meaningful formative comment, therefore, they want the project to remain a part of their math education training.

This result affirms the importance of a closer contact between pre-service teachers and schoolchildren. The project CAMI can be seen as one of possible ways of such a practice-oriented training, although some students would be more interested in getting a more direct contact with schoolchildren in the real classroom. We can conclude that there is a need for more variety in projects of mathematics education courses.

The next issue coming from our findings is related to the problem of differentiation in the mathematics classroom. As we said in our previous sections, today's mathematics curriculum recognizes differences in how children learn and the right of all children to receive an education that is adapted to their needs. The New Brunswick math curriculum stresses that in order to meet educational need of each student; teachers have to use a variety of approaches. CAMI presents one possible resource that gives pupils a chance to choose an appropriate problem, solve it at their own pace using their own strategy and communication tool. It brings also some informal and challenging elements in the classroom routine. The fact that each participant gets a personal comment from a university student can be seen as a motivating factor for schoolchildren because they see that their work brings the attention of other people and is being socially valorized by a personal attention or even a public recognition (children can see this recognition when their solution is posted as interesting or their name is placed on the congratulation list). However, more research is needed in order to study in more detail the factors of intrinsic motivation among participants of the CAMI project.

Our further plan is to continue the development of the CAMI community adding new functions and new problems. We are planning to find technical and pedagogical solutions on how to give a better and faster feedback to the children and make the site more dynamic and attractive. We will encourage all participants to contribute to the CAMI bank of problems adding more problems composed by schoolchildren, their teachers and university students. We are thinking also to add a science and chess section to our site and we are open to international collaborations with different institutions to work together on this project.

9. References

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